

Chapter 7 Multiple Regression Analysis with Qualitative Information

- 1 Using the data in SLEEP75.RAW (see also Problem 3 in Chapter 3), we obtain the estimated equation

$$\begin{aligned}\widehat{sleep} &= 3,840.83 - .163 \text{ totwrk} - 11.71 \text{ educ} - 8.70 \text{ age} \\ &\quad (235.11) \quad (.018) \quad (5.86) \quad (11.21) \\ &\quad + .128 \text{ age}^2 + 87.75 \text{ male} \\ &\quad (.134) \quad (34.33) \\ n &= 706, R^2 = .123, \bar{R}^2 = .117.\end{aligned}$$

The variable *sleep* is total minutes per week spent sleeping at night, *totwrk* is total weekly minutes spent working, *educ* and *age* are measured in years, and *male* is a gender dummy.

- (i) All other factors being equal, is there evidence that men sleep more than women? How strong is the evidence?
- (ii) Is there a statistically significant tradeoff between working and sleeping? What is the estimated tradeoff?
- (iii) What other regression do you need to run to test the null hypothesis that, holding other factors fixed, age has no effect on sleeping?

Suggested Answer

(i) The coefficient on *male* is 87.75, so a man is estimated to sleep almost one and one-half hours more per week than a comparable woman. Further, $t_{male} = 87.75/34.33 \approx 2.56$, which is close to the 1% critical value against a two-sided alternative (about 2.58). Thus, the evidence for a gender differential is fairly strong.

(ii) The t statistic on *totwrk* is $-.163/.018 \approx -9.06$, which is very statistically significant. The coefficient implies that one more hour of work (60 minutes) is associated with $.163(60) \approx 9.8$ minutes less sleep.

(iii) To obtain R_r^2 , the R -squared from the restricted regression, we need to estimate the model without *age* and *age*². When *age* and *age*² are both in the model, *age* has no effect only if the parameters on both terms are zero.

- 2 The following equations were estimated using the data in BWGHT.RAW:

$$\begin{aligned}\widehat{\log(bwght)} &= 4.66 - .0044 \text{ cigs} + .0093 \log(\text{faminc}) + .016 \text{ parity} \\ &\quad (.22) \quad (.0009) \quad (.0059) \quad (.006) \\ &\quad + .027 \text{ male} + .055 \text{ white} \\ &\quad (.010) \quad (.013) \\ n &= 1,388, R^2 = .0472\end{aligned}$$

and

$$\begin{aligned}\widehat{\log(bwght)} = & 4.65 - .0052 \text{ cigs} + .0110 \log(\text{faminc}) + .017 \text{ parity} \\ & (.38) \quad (.0010) \quad (.0085) \quad (.006) \\ & + .034 \text{ male} + .045 \text{ white} - .0030 \text{ motheduc} + .0032 \text{ fatheduc} \\ & (.011) \quad (.015) \quad (.0030) \quad (.0026) \\ n = & 1,191, R^2 = .0493.\end{aligned}$$

The variables are defined as in Example 4.9, but we have added a dummy variable for whether the child is male and a dummy variable indicating whether the child is classified as white.

- (i) In the first equation, interpret the coefficient on the variable *cigs*. In particular, what is the effect on birth weight from smoking 10 more cigarettes per day?
- (ii) How much more is a white child predicted to weigh than a nonwhite child, holding the other factors in the first equation fixed? Is the difference statistically significant?
- (iii) Comment on the estimated effect and statistical significance of *motheduc*.
- (iv) From the given information, why are you unable to compute the *F* statistic for joint significance of *motheduc* and *fatheduc*? What would you have to do to compute the *F* statistic?

Suggested Answer

- (i) 4.4% less
- (ii) 5.5% more, *t* test finds that it is statistically significant
- (iii) Insignificant
- (iv) Number of observations differs in two regressions. Keep the observations in common.

3 Using the data in GPA2.RAW, the following equation was estimated:

$$\begin{aligned}\widehat{\text{sat}} = & 1,028.10 + 19.30 \text{ hsize} - 2.19 \text{ hsize}^2 - 45.09 \text{ female} \\ & (6.29) \quad (3.83) \quad (.53) \quad (4.29) \\ & - 169.81 \text{ black} + 62.31 \text{ female} \cdot \text{black} \\ & (12.71) \quad (18.15) \\ n = & 4,137, R^2 = .0858.\end{aligned}$$

The variable *sat* is the combined SAT score, *hsize* is size of the student's high school graduating class, in hundreds, *female* is a gender dummy variable, and *black* is a race dummy variable equal to one for blacks and zero otherwise.

- (i) Is there strong evidence that *hsize*² should be included in the model? From this equation, what is the optimal high school size?
- (ii) Holding *hsize* fixed, what is the estimated difference in SAT score between nonblack females and nonblack males? How statistically significant is this estimated difference?
- (iii) What is the estimated difference in SAT score between nonblack males and black males? Test the null hypothesis that there is no difference between their scores, against the alternative that there is a difference.
- (iv) What is the estimated difference in SAT score between black females and nonblack females? What would you need to do to test whether the difference is statistically significant?

Suggested Answer

(i) The t statistic on $hsize^2$ is over four in absolute value, so there is very strong evidence that it belongs in the equation. We obtain this by finding the turnaround point; this is the value of $hsize$ that maximizes $\hat{s}at$ (other things fixed): $19.3/(2 \cdot 2.19) \approx 4.41$. Because $hsize$ is measured in hundreds, the optimal size of graduating class is about 441.

(ii) This is given by the coefficient on *female* (since *black* = 0); nonblack females have SAT scores about 45 points lower than nonblack males. The t statistic is about -10.51 , so the difference is very statistically significant. (The very large sample size certainly contributes to the statistical significance.)

(iii) Because *female* = 0, the coefficient on *black* implies that a black male has an estimated SAT score almost 170 points less than a comparable nonblack male. The t statistic is over 13 in absolute value, so we easily reject the hypothesis that there is no *ceteris paribus* difference.

(iv) We plug in *black* = 1, *female* = 1 for black females and *black* = 0 and *female* = 1 for nonblack females. The difference is therefore $-169.81 + 62.31 = -107.50$. Because the estimate depends on two coefficients, we cannot construct a t statistic from the information given. The easiest approach is to define dummy variables for three of the four race/gender categories and choose nonblack females as the base group. We can then obtain the t statistic we want as the coefficient on the black female dummy variable.

- 6 To test the effectiveness of a job training program on the subsequent wages of workers, we specify the model

$$\log(\text{wage}) = \beta_0 + \beta_1 \text{train} + \beta_2 \text{educ} + \beta_3 \text{exper} + u,$$

where *train* is a binary variable equal to unity if a worker participated in the program. Think of the error term u as containing unobserved worker ability. If less able workers have a greater chance of being selected for the program, and you use an OLS analysis, what can you say about the likely bias in the OLS estimator of β_1 ? (Hint: Refer back to Chapter 3.)

Suggested Answer

$$\log(\widehat{\text{wage}}) = \hat{\beta}_0 + \hat{\beta}_1 \text{train} + \hat{\beta}_2 \text{educ} + \hat{\beta}_3 \text{exper} + \hat{\beta}_4 \text{abil} + \mu$$

$$\log(\widetilde{\text{wage}}) = \tilde{\beta}_0 + \tilde{\beta}_1 \text{train} + \tilde{\beta}_2 \text{educ} + \tilde{\beta}_3 \text{exper} + \mu$$

$$\tilde{\beta}_1 = \hat{\beta}_1 + \hat{\beta}_4 \delta_1$$

In which δ_1 is the coefficient from

$$\widetilde{\text{abil}} = \delta_1 \text{train} + \delta_2 \text{educ} + \delta_3 \text{exper} + \varepsilon$$

$$\hat{\beta}_4 > 0$$

$$\tilde{\delta}_1 < 0$$

$$\hat{\beta}_4 \delta_1 < 0$$

Downward bias